OliveTrack Data Architecture Reference

Complete field dictionary accompanying the working specification

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1 Complete Field Dictionary

This inventory reflects the consolidated business-rule review. All business tables use stable UUID identities and same-workspace references. Accepted-record requiredness is separate from local pending state. Common metadata includes creation/update time and revision, not a permanent full field-change history. Technical authorisation/outbox records are separate from the 34 domain entities/associations, including explicit WorkerAvailability.

1.1 User

`app_user`. UNIQUE(normalised_email); authentication subject uniqueness where applicable. No global business role.

Field / type	Presence	Meaning
`name` `text`	Required	Account holder name.
`email` `text`	Required	Email as presented by the account holder.
`normalised_email` `text`	Required	Canonical identity value using the agreed normalisation policy.
`account_status` `text`	Required	Pending verification or active under the defined lifecycle.
`email_verified_at` `timestamptz`	Optional	Successful email verification time.
`auth_reference` `text`	Conditional	Reference to the selected identity mechanism, if external to this record.

1.2 Workspace

`workspace`. Owner is derived from workspace_membership, not duplicated independently on properties.

Field / type	Presence	Meaning
`name` `text`	Required	Owner business workspace name.
`status` `text`	Required	Technical availability of the workspace; active creation requires an Owner membership.
`business_timezone` `text`	Required	Configured IANA business timezone used for event date association and display.

1.3 WorkspaceMembership

`workspace_membership`. UNIQUE(workspace_id,user_id); at most one Owner per workspace and at most one owned workspace per User. Active workspaces require an Owner transactionally.

Field / type	Presence	Meaning
`user_id` `uuid`	Required	Account belonging to this workspace.
`workspace_role` `text`	Required	Owner or Field Manager.
`access_status` `text`	Required	Active or revoked membership.
`revoked_at` `timestamptz`	Conditional	Time of revocation for a manager membership.

1.4 ManagerInvitation

`manager_invitation`. Included invitation feature. Persist no plaintext acceptance secret; invitation token implementation follows the selected authentication design.

Field / type	Presence	Meaning
`recipient_email` `text`	Required	Intended recipient identity under the common normalisation policy.
`invited_by_user_id` `uuid`	Required	Owner issuing the invitation.
`invitation_status` `text`	Required	Pending, accepted, cancelled or expired.
`expires_at` `timestamptz`	Required	Expiry under the agreed token policy.
`accepted_membership_id` `uuid`	Conditional	Membership created or restored on acceptance.
`proposed_access` `jsonb`	Required	Reviewed initial property/operation grants and explicitly available workers; validate workspace and targets at invitation and acceptance.

1.5 AccessGrant

`access_grant`. UNIQUE(membership_id, operation_code, property_id). Property scope is required and includes parcels and current/future campaigns until revoked; no separate campaign/parcel grants. Multiple managers may share a property.

Field / type	Presence	Meaning
`membership_id` `uuid`	Required	Same-workspace manager membership.
`operation_code` `text`	Required	Independent consultation, entry, correction or specialised operation.

Field / type	Presence	Meaning
`property_id` `uuid`	Required	Same-workspace property; no scope-kind/campaign-target alternative.

Filter campaign results by authorised property contributions; do not reveal full mixed-property records/files outside that scope. Worker relevance grants no access to other-property work. Daily attendance remains a single workspace/worker/date state; overlapping manager edits use revision/conflict resolution.

1.6 WorkerAvailability

`worker_availability`. UNIQUE(membership_id, worker_id). Owner-explicit availability before scoped plans/work exist; same-workspace FKs. This association grants no additional operation or other-property work access.

Field / type	Presence	Meaning
`membership_id` `uuid`	Required	Manager allowed to select this worker.
`worker_id` `uuid`	Required	Worker in the same business.

1.7 Property

`property`. Parcel and all operational references must remain in the same workspace; stable identity survives descriptive edits.

Field / type	Presence	Meaning
`name` `text`	Required	Property name.
`textual_location` `text`	Required	Location description without requiring a map.
`declared_area_ha` `numeric`	Optional	Owner-supplied reference area in hectares; unknown remains NULL.
`olive_varieties` `text[]`	Optional	Variety names; a controlled registry is not assumed.
`is_archived` `boolean`	Required	Property availability flag.

1.8 Parcel

`parcel`. Parent workspace consistency. No invented uniqueness of names or requirement for parcel areas to total the property area.

Field / type	Presence	Meaning
property_id uuid	Required	Parent property.
name text	Required	Named property division.
declared_area_ha numeric	Optional	Declared parcel area in hectares.
olive_varieties text[]	Optional	Known olive varieties.
is_archived boolean	Required	Independent parcel availability flag.

1.9 HarvestCampaign

harvest_campaign. At least one coverage entry. Multiple Planned campaigns may share areas. Transactional exclusive execution checks at start/reopen/coverage change; preserve valid actual chronology and independent archival.

Field / type	Presence	Meaning
name text	Required	Campaign name.
planned_start_date date	Required	Expected start, separate from actual events.
actual_start_date date	Conditional	Explicitly confirmed actual start; required in accepted In progress/Closed. Offline proposal is not central acceptance.
actual_end_date date	Conditional	Actual end required for closed campaigns.
lifecycle_status text	Required	Planned, In progress or Closed. Start: Planned -> In progress; reopen: Closed -> In progress. Only In progress reserves execution.
is_archived boolean	Required	Visibility attribute independent of lifecycle.

1.10 CampaignCoverage

campaign_coverage. Unique campaign/property/parcel selection with explicit whole-property NULL handling. No parent/child duplicate in one campaign. Conflicting coverage is restricted for accepted In progress execution, not Planned preparation.

Field / type	Presence	Meaning
campaign_id uuid	Required	Covered campaign.
property_id uuid	Required	Covered property or parent of the selected parcel.

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Field / type	Presence	Meaning
parcel_id uuid	Optional	NULL means whole property; otherwise the selected parcel.

1.11 Worker

worker. Stable client-generatable UUID primary key; non-unique business/type/country/normalised-number and phone/name lookups for duplicate warnings. No automatic cross-business merge or mandatory document-uniqueness rejection. updated_at retains current profile modification time.

Field / type	Presence	Meaning
full_name text	Required	Worker name within the business.
telephone text	Required	Contact number; not a globally unique identity.
official_id_type text	Required	Official identification type; accepted types remain to be confirmed with the client.
official_id_country text	Required	Issuing country; country coding and format validation remain implementation details.
official_id_number text	Required	Official identification number as text; not the internal primary key or proof of verification.
can_operate_machinery boolean	Required	Recorded capability, separate from actual task.
capability_description text	Optional	Additional capability description.
is_active boolean	Required	New-selection availability.

1.12 WorkPlan

work_plan. UNIQUE(workspace_id,campaign_id,work_date). Accepted plan contains assignments; multi-day creation produces independent daily plans.

Field / type	Presence	Meaning
campaign_id uuid	Required	Planned campaign.
work_date date	Required	Target working date.

1.13 PlannedAssignment

`planned_assignment`. Unique plan/worker/area, with explicit NULL handling. Eligibility and required periods for multiple areas are cross-record rules.

Field / type	Presence	Meaning
`plan_id` `uuid`	Required	Daily campaign plan.
`worker_id` `uuid`	Required	Expected worker.
`property_id` `uuid`	Required	Planned property.
`parcel_id` `uuid`	Optional	Planned parcel within that property.

1.14 PlannedWorkPeriod

`planned_work_period`. Positive intervals and no defined planned overlap for the same worker across campaigns. An assignment with unknown hours has no invented period.

Field / type	Presence	Meaning
`assignment_id` `uuid`	Required	Source planned assignment.
`worker_id` `uuid`	Required	Denormalised worker identity checked against the assignment for conflict enforcement.
`planned_start` `timestamptz`	Required	Complete planned start.
`planned_end` `timestamptz`	Required	Complete planned end after start.
`task_code` `text`	Optional	Machinery operation or general work; unknown is NULL.

1.15 PlannedBreak

`planned_break`. Positive interval. Break validation follows Q-WF-03; no second deduction for already excluded time.

Field / type	Presence	Meaning
`assignment_id` `uuid`	Required	Planned worker/area context.
`break_start` `timestamptz`	Required	Proposed break start.
`break_end` `timestamptz`	Required	Proposed break end.

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Field / type	Presence	Meaning
counts_as_work boolean	Required	Proposed treatment, subject to actual confirmation.

1.16 DailyAttendance

daily_attendance. UNIQUE(workspace_id,worker_id,work_date). No campaign_id: daily attendance is shared across campaign contexts.

Field / type	Presence	Meaning
worker_id uuid	Required	Worker attending in this workspace.
work_date date	Required	Start-based business working date.
attendance_status text	Required	Awaiting confirmation only for selected/expected workers, or explicitly Present/Absent. Unrelated unselected workers need no row; one workspace/worker/start-based working date across areas.

1.17 ActualWorkAssignment

actual_work_assignment. Unique attendance/campaign/area with NULL handling. Presence, campaign coverage and lifecycle must be consistent; periods may be absent.

Field / type	Presence	Meaning
attendance_id uuid	Required	Daily attendance providing the worker and working date.
worker_id uuid	Required	Worker identity, validated against attendance for period constraints.
campaign_id uuid	Required	Actual campaign.
property_id uuid	Required	Actual property.
parcel_id uuid	Optional	Actual parcel.
source_planned_ assignment_id uuid	Optional	Proposal used as a source, without requiring a one-to-one outcome.

1.18 WorkPeriod

work_period. Positive, non-overlapping actual periods for the same worker across all campaign contexts. Incomplete hours remain at assignment level.

Field / type	Presence	Meaning
assignment_id uuid	Required	FK to ActualWorkAssignment (actual_work_assignment.id), not to PlannedAssignment.
worker_id uuid	Required	Must match the worker of the referenced ActualWorkAssignment; used for overlap checks.
actual_start timestamptz	Required	Complete actual start, defining the working date.
actual_end timestamptz	Required	Complete actual end after start.
task_code text	Required	Actual task explicitly selected for the complete confirmed period; profile machinery capability does not determine it.

1.19 WorkBreak

work_break. Use complete intervals and the same-worker union/intersection calculation. Overlapping/out-of-period entry handling remains under Q-WF-03.

Field / type	Presence	Meaning
attendance_id uuid	Required	Daily attendance identifying the worker and working-date context.
break_start timestamptz	Required	Actual break start.
break_end timestamptz	Required	Actual break end.
counts_as_work boolean	Required	Only false reduces intersecting actual time.

1.20 Delivery

delivery. Accepted departure has exactly one transport and at least one batch. Correction, cancellation and reassignment follow D-13 (resolves Q-DA-06/Q-DA-07): require separate authority, reopen closed affected campaigns and validate all dependencies. Cancellation retains sources and excludes their totals; deletion requires no external dependants.

Field / type	Presence	Meaning
campaign_id uuid	Required	Single source campaign.
destination_name text	Required	Receiving destination, reused or entered directly.

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Field / type	Presence	Meaning
delivery_status text	Required	In transport, delivered or cancelled. Cancellation preserves linked source records and excludes their totals.
delivered_at timestamptz	Conditional	Required when delivered, not before departure.
cancellation_reason text	Conditional	Required when an erroneous delivery record is cancelled.
cancelled_at timestamptz	Conditional	Accepted cancellation time; not a physical journey event.
cancelled_by_user_id UUID FK	Conditional	Authorised cancelling account; local origin/acceptance metadata stays distinct.

1.21 TransportRecord

transport_record. UNIQUE(delivery_id) implements the current one-associated-transport model under D-13; representative transport examples remain under Q-DA-10. No independent trip workflow.

Field / type	Presence	Meaning
delivery_id uuid	Required	Associated delivery.
driver_worker_id uuid	Required	Existing same-business worker under the current proposal.
vehicle_reference text	Required	Vehicle identifier or reference.
departure_at timestamptz	Required	Actual departure time.
transport_document_ reference text	Optional	Reported accompanying transport-document reference.
declared_origin_ property_id UUID FK	Optional	Property declared in the transport document; separate from actual source batches.
declared_origin_parcel_ id UUID FK	Optional	Declared parcel, if known and represented in the business; belongs to the declared property.
declared_parcel_ reference text	Optional	External parcel identifier as reported; does not require all properties to be divided into parcels.

1.22 OliveBatch

`olive_batch`. Source areas must be eligible. Different origin/variety parts from the same area are allowed; do not add a simplistic UNIQUE(delivery_id,property_id). Represents source quantities, not source monetary amounts; mixed-load amounts remain unallocated by area.

Field / type	Presence	Meaning
`delivery_id` `uuid`	Required	Physical delivery containing this part of the load.
`property_id` `uuid`	Required	Source property in campaign coverage.
`parcel_id` `uuid`	Optional	Source parcel.
`collection_origin` `text`	Optional	Tree-collected or ground-collected when known.
`olive_variety` `text`	Optional	Known variety for this load part.
`harvesting_method` `text`	Optional	Known method, distinct from collection origin.
`quantity_value` `numeric`	Optional	Known source quantity, not inferred from the ticket total.
`quantity_unit` `text`	Conditional	Required when quantity is supplied.
`quantity_meaning` `text`	Conditional	Measurement basis needed before aggregation.

1.23 WeighingTicket

`weighing_ticket`. UNIQUE(delivery_id, workspace_id); one ticket per delivery. Ticket reference remains source-contextual. Clean net weight is not duplicated in reported_weights.

Field / type	Presence	Meaning
`delivery_id` `uuid`	Required	Exactly one associated delivery.
`ticket_reference` `text`	Optional	Source identifier, not assumed globally unique.
`cooperative_name` `text`	Optional	Reporting entity when available.
`weighed_at` `timestamptz`	Optional	Source weighing time.
`reported_weights` `jsonb`	Optional	Schema-validated source entries for other reported weights: value, unit, source label, meaning and interpretation status. The clean net kg value has its own canonical column.

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Field / type	Presence	Meaning
`interpretation_note` `text`	Optional	Ambiguous or incomplete source interpretation.
`clean_net_weight_kg` `numeric`	Optional	Reported clean net weight in kg, canonical basis for the delivery yield weighting; not derived from unspecified deductions.
`weighing_started_at` `timestamptz`	Optional	Reported weighing start instant.
`weighing_ended_at` `timestamptz`	Optional	Reported weighing end, not before known start.
`excess_time_value` `numeric`	Optional	Reported excess duration; never calculated from an assumed scale rate.
`excess_time_unit` `text`	Conditional	Required with excess duration, e.g. seconds or minutes.

1.24 TicketAdjustment

`ticket_adjustment`. No deduction formula or ordering is assumed. Monetary adjustments require financial authorisation and are excluded from offline copies.

Field / type	Presence	Meaning
`ticket_id` `uuid`	Required	Owning ticket.
`adjustment_type` `text`	Optional	Reported adjustment type, including excess-time monetary penalty; financial permission applies to monetary records.
`reported_value` `numeric`	Optional	Reported adjustment value.
`unit` `text`	Optional	Reported unit, such as percentage, mass or monetary unit.
`calculation_basis` `text`	Optional	Known basis and source interpretation.
`already_applied` `boolean`	Optional	True, false or unknown; NULL must not be treated as false.

1.25 YieldAnalysis

`yield_analysis`. UNIQUE(delivery_id, workspace_id). Reference weight is read from this delivery's single WeighingTicket.clean_net_weight_kg; no independently editable copy.

Field / type	Presence	Meaning
`delivery_id` `uuid`	Required	Exactly one analysed delivery.

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Field / type	Presence	Meaning
`analysis_reference` `text`	Optional	Source reference.
`reporting_entity` `text`	Optional	Cooperative or laboratory.
`analysis_date` `date`	Optional	Source analysis date, separate from entry and delivery dates.
`yield_percentage` `numeric`	Optional	Reported result; missing is not zero.

1.26 TicketValuation

`ticket_valuation`. UNIQUE(ticket_id). Update the current row when final replaces provisional; no estimates/history rows. Final-to-provisional policy remains Q-DA-09.

Field / type	Presence	Meaning
`ticket_id` `uuid`	Required	Valued ticket, not an independent delivery amount.
`amount` `numeric`	Required	Current reported amount with two decimal places; missing valuation has no invented row.
`currency` `text`	Required	EUR in the current scope.
`valuation_status` `text`	Required	Provisional or final.
`reporting_cooperative` `text`	Required	Source, normally obtained from reviewed ticket context.
`reported_date` `date`	Required	Date the amount was communicated.
`source_reference` `text`	Optional	Supporting reference.
`observations` `text`	Optional	Additional current information.

1.27 Document

`document`. Accepted document has worker association(s) and version(s). Current version is derived from accepted version order, not from validity or signature state.

Field / type	Presence	Meaning
`title` `text`	Required	Document title.
`document_category` `text`	Required	Contract, insurance or other.

1.28 DocumentWorker

`document_worker`. UNIQUE(document_id,worker_id). Association changes cannot invalidate retained signer evidence; workers are not deleted while dependent records exist.

Field / type	Presence	Meaning
`document_id` `uuid`	Required	Shared document identity.
`worker_id` `uuid`	Required	Covered worker in the same business.

1.29 DocumentVersion

`document_version`. UNIQUE(document_id,version_number). New file creates a new version. Current is the highest accepted version number, including if withdrawn; previous is the immediately preceding accepted version. Retain all signed versions and externally declared signed uploads. Protect pending signing operations known to the server; preserve exclusively local signatures for Q-DOC-04 resolution if their source becomes unavailable; only eligible older unsigned non-contract versions may be cleaned up.

Field / type	Presence	Meaning
`document_id` `uuid`	Required	Versioned document.
`version_number` `integer`	Required	Positive monotonically allocated version sequence.
`source_file_id` `uuid`	Required	Available StoredFile containing the accepted source contents.
`uploaded_by_user_id` `uuid`	Required	Responsible account.
`uploaded_at` `timestamptz`	Required	Accepted upload timestamp.
`insurance_expiry_date` `date`	Optional	Expiry when known; only meaningful for insurance category.
`declared_signed_externally` `boolean`	Required	Explicit declaration for this version, defaulting to no declaration.
`is_withdrawn` `boolean`	Required	Operational availability, independent of currency/validity/signature.
`withdrawn_at` `timestamptz`	Conditional	Withdrawal event timestamp when applicable.
`origin_kind` `text`	Required	Uploaded or generated; both use the same version/signature lifecycle.
`template_id` `UUID FK`	Conditional	Template used for generated source versions; absent for uploads.
`template_revision` `text`	Conditional	Frozen reference to the template revision used at generation.

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Field / type	Presence	Meaning
`generation_inputs` `jsonb`	Conditional	Reviewed snapshot of generation fields, not live worker profile values. Protected with document permissions.

1.30 WorkerSignature

`worker_signature`. UNIQUE(worker_id,version_id) for accepted results. Pending local results are operations until accepted. A different confirmed drawing never replaces this pair.

Field / type	Presence	Meaning
`worker_id` `uuid`	Required	The actual signer, associated with the document.
`version_id` `uuid`	Required	The reviewed version.
`initiated_by_user_id` `uuid`	Required	Authorised Owner/manager facilitating signing, not an alternative signer.
`client_signed_at` `timestamptz`	Required	Recorded signing time supplied by the device; not automatically trusted as server time.
`server_accepted_at` `timestamptz`	Required centrally	Authoritative acceptance time, separate from signing time.
`signature_file_id` `uuid`	Required in this proposal	Stored representation of the confirmed drawn signature.

1.31 StoredFile

`stored_file`. UNIQUE(storage_key). Enforce workspace, responsibility scope and complete-file consultation through the associated business context. Manager financial consultation starts enabled; source upload and structured financial entry/correction remain separate operations. No mandatory monetary-content classification.

Field / type	Presence	Meaning
`storage_key` `text`	Required	Opaque reference in the selected object/file store.
`original_filename` `text`	Required	Supplied filename, not access authority.
`media_type` `text`	Required	Validated file format.
`byte_size` `bigint`	Required	Measured bytes; initial upload limit is 10 MB, with exact byte convention pending.
`content_hash` `text`	Required for accepted document source	Cryptographic digest under SEC-08; protect metadata and verify source integrity. Algorithm/implementation remain architecture choices. Other file categories follow their integrity policy.

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Field / type	Presence	Meaning
storage_status text	Required	Technical uploading, available or deletion-pending state; not a document approval stage.

1.32 SourceAttachment

source_attachment. Exactly one target FK is populated; all targets/files share the workspace. Separate source-attachment policies under Q-AT-02, not automatic document-version retention.

Field / type	Presence	Meaning
stored_file_id uuid	Required	Stored source evidence.
ticket_id uuid	Alternative	Ticket target.
analysis_id uuid	Alternative	Analysis target.
valuation_id uuid	Alternative	Current valuation target.
uploaded_by_user_id uuid	Required	Account supplying the evidence.

1.33 DocumentTemplate

document_template. Template and source file share the workspace. Existing output keeps a reviewed input/revision snapshot when template availability changes.

Field / type	Presence	Meaning
name text	Required	Template display name.
document_category text	Required	Contract, insurance or other category of generated output.
template_file_id UUID FK	Required	Private template contents in StoredFile.
revision text	Required	Identifies the template revision used by accepted outputs.
field_schema jsonb	Required	Validated list of prefillable/required fields and supported single/shared-worker rendering.
is_available boolean	Required	Whether selectable for new generation; not a template-management user workflow.

1.34 DocumentRendition

`document_rendition`. UNIQUE(version_id, worker_id, workspace_id); output requires the accepted worker/version signature centrally. Pending local outputs remain proposals. Preserve signed source/evidence/rendition together.

Field / type	Presence	Meaning
`version_id` `UUID FK`	Required	Reviewed source version; never overwritten by a rendition.
`worker_id` `UUID FK`	Required	Associated worker whose application signature appears in this output.
`stored_file_id` `UUID FK`	Required	Derived worker-specific PDF contents.
`generated_at` `timestamptz`	Required	Rendition generation time; not substituted for source signing time.